All prompts — the full comparison

Every prompt we tried for MedGemma-27B, and how it did. All numbers are **pooled over 1994 reports** (Zoe n=1495 + Maria n=499), **Core F1** and whole-report accuracy vs the reference annotator **LD**, with the second annotator **SG** held out. Baselines: the paper's **Mistral-7B** (Tian et al.) and the **human** second annotator (SG vs LD).

The prompts

Prompt	Base	What it adds (technical)	Outcome
v1	—	Impression-first; short ACNS-style definitions; 5-label 1–4 schema; JSON output under a GBNF grammar	Baseline — 83.9% whole; already beats Mistral on 4/5 categories
v2	v1	Neurologist *system* role; read Description/body **first**; keep body findings over a conservative Impression; extended ACNS/ILAE definitions	Helps Maria abnormality but **over-calls Focal Epi** → weaker overall
v3	v1	Explicit **focal-epileptiform exclusion list**: generalized/bilateral discharges, "sharply-contoured"/benign variants, artifacts, and focal slowing all ≠ focal-epi	↑ Focal-Epi precision without losing recall — **kept**
v4	v1	Step-by-step epileptiform **procedure**: detect → localize (focal/generalized) → assign	≈ v3, marginally worse
v5	v3	**Focal-vs-generalized discriminator for slowing** (read distribution from the wording; forbid double-flagging one finding)	↑ the slowing classes — **best prompt content**
v6	v3	Prompt asks the model to **reconcile** overall/subtype consistency itself	**Rejected** — asking does not hold; even hurts other fields
v7	v5	**Body-aware abnormality** — call it abnormal on a body finding the Impression downplays	**Rejected** — no gain on Abnormality
v8	—	Deliberately **simplified** (~340 tokens): only the two decision questions, no exclusion lists	**Rejected** — Focal-Epi collapses (exclusions are load-bearing)

Prompt	Base	What it adds (technical)	Outcome
v9	—	**Reasoning-first**: concise frame + a free-text `reasoning` field before the labels (larger max-tokens)	**Rejected** — 3–6 pts worse, much slower
v10	v5	**Evidence-calibration** rule: tie confidence to explicit textual support; prefer *absent* when balanced	**Rejected** — ~1 pt below v5

A **…g** suffix (e.g. **v5g**) means the variant is run with **grammar-enforced consistency** (`ENFORCE_CONSISTENCY=1`): a GBNF grammar emits `overall_abnormal` last and only lets it be abnormal when a subtype is, so a self-contradictory answer is impossible to produce. Full prompt texts are in `../prompts/`.

Where every prompt lands

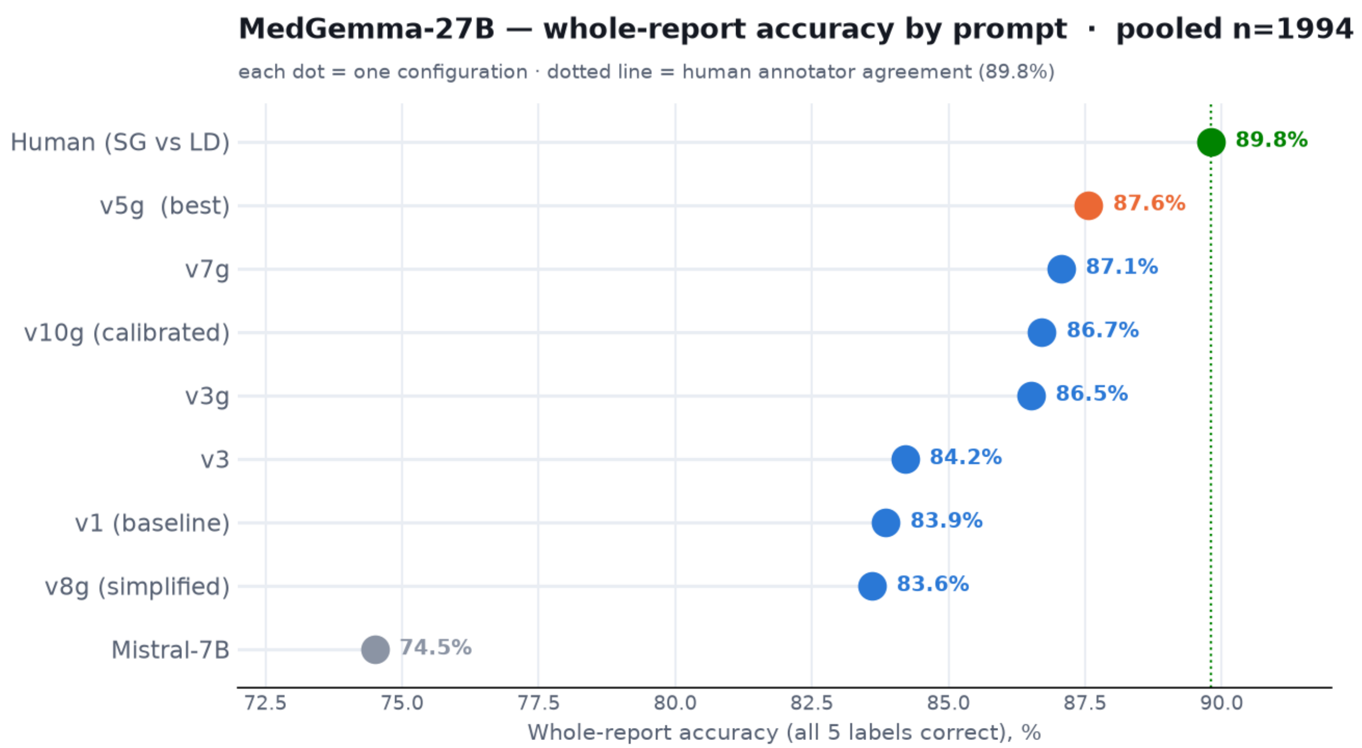

Figure 1: Whole-report accuracy by prompt

What you see: whole-report accuracy (all five labels correct) for each configuration, from Mistral-7B up to the human agreement line (89.8%, dotted). Our baseline **v1 already clears Mistral by ~10 points.** Adding grammar-enforced consistency is the big step (**v3g/v5g**), and **v5g (87.6%)** is the best — within ~2 points of the human ceiling. The rejected ideas cluster below it: **v8 (simplified)** even falls *below* the v1 baseline, because dropping the detailed rules brings the over-calling straight back.

Our top prompts vs Mistral and human, per category

What you see: Core F1 for **each label scored on its own**, for our three strongest prompts (**v5g**, **v3g**, **v7g**), Mistral-7B, and the human annotator. Note these are a *different metric* from the first chart:

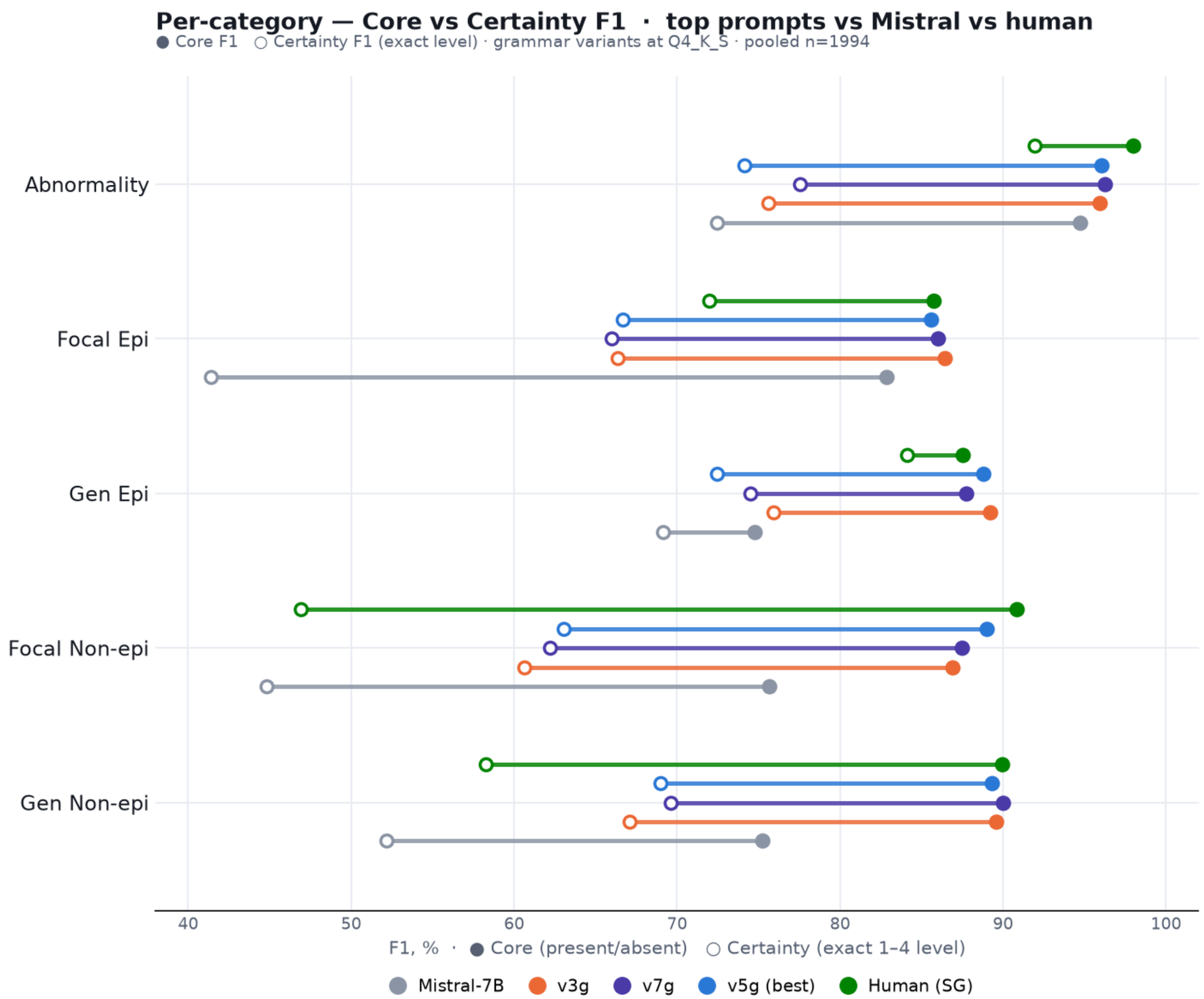

Figure 2: Per-category F1 — top prompts vs Mistral and human

the first chart's 87.6% is the share of reports with **all five** labels correct at once; here each number is the F1 of **one** label — so the two are not meant to match. (Grammar variants shown at **Q4_K_S** — the same quant as the Full-numbers table, so the dots match those rows exactly.) Our prompts (blue / orange / violet) **cluster tightly together and sit far above Mistral (grey)** on the harder three classes (Gen Epi, Focal Non-epi, Gen Non-epi) — the gap to Mistral is 13–14 points there. Against the **human (green)** the cluster sits right at the human level: **at or slightly above it on the epileptiform classes** (Focal Epi ~86 vs 86, Gen Epi ~88–89 vs 88) and on **Gen Non-epi** (~89–90 vs 90), trailing only on **Abnormality** and **Focal Non-epi** — where even the two humans disagree most. That the three prompts land so close to each other is itself the point: the result is stable across our top variants, not a fluke of one prompt.

Why we win — confidence, not just direction

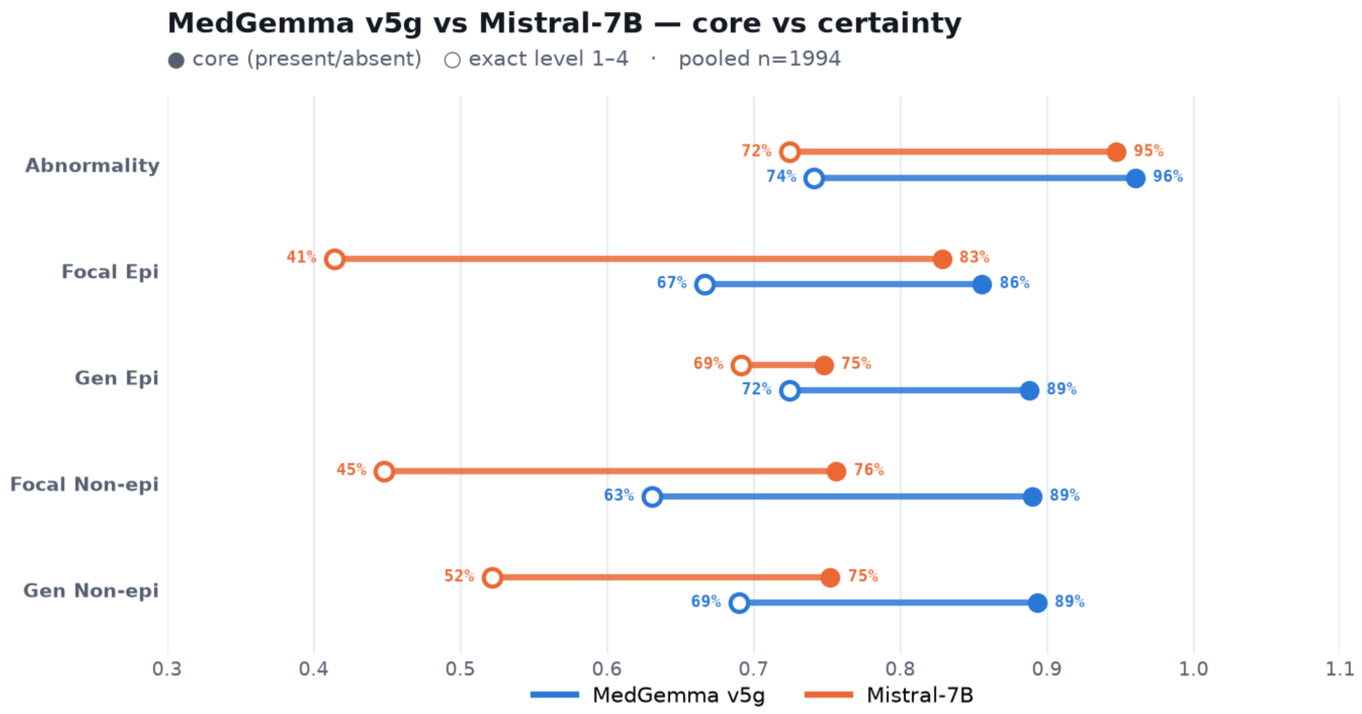

Figure 3: v5g vs Mistral — core vs certainty

What you see: each line runs from **Core F1** (● — did we get present/absent right?) to **Certainty F1** (○ — did we get the *exact* 1–4 confidence level?); the line length is how much is lost when the exact level is required. Our **v5g (blue)** beats Mistral on core almost everywhere, but the real separation is the **exact level**: Mistral's open circles collapse (Focal Epi 83 → **41**, Focal Non-epi 76 → 45, Gen Non-epi 75 → 52) — it points in the right direction but badly misjudges *how sure* to be — while v5g holds far better (Focal Epi 86 → 67, Focal Non-epi 89 → 63). Getting the confidence level right, not just the yes/no, is where our prompt+grammar approach pulls ahead.

Which quantization, and does it generalize?

What you see: v5g on the two model sizes. They tie on whole-report accuracy (87.6% each); per category **Q2_K** is stronger on the epileptiform classes (Focal Epi 88 vs 86) while **Q4_K_S** is stronger on the diffuse/slowing classes (Gen Non-epi 89 vs 87). Pick Q2_K for half the footprint, Q4_K_S if encephalopathy/diffuse findings matter most.

What you see: the same model on the **seen** neurologist (Zoe) and an **unseen** one (Maria). Performance holds across reporting styles — the epileptiform and focal classes are as strong or stronger on the unseen data — so the model generalizes rather than fitting one annotator's phrasing.

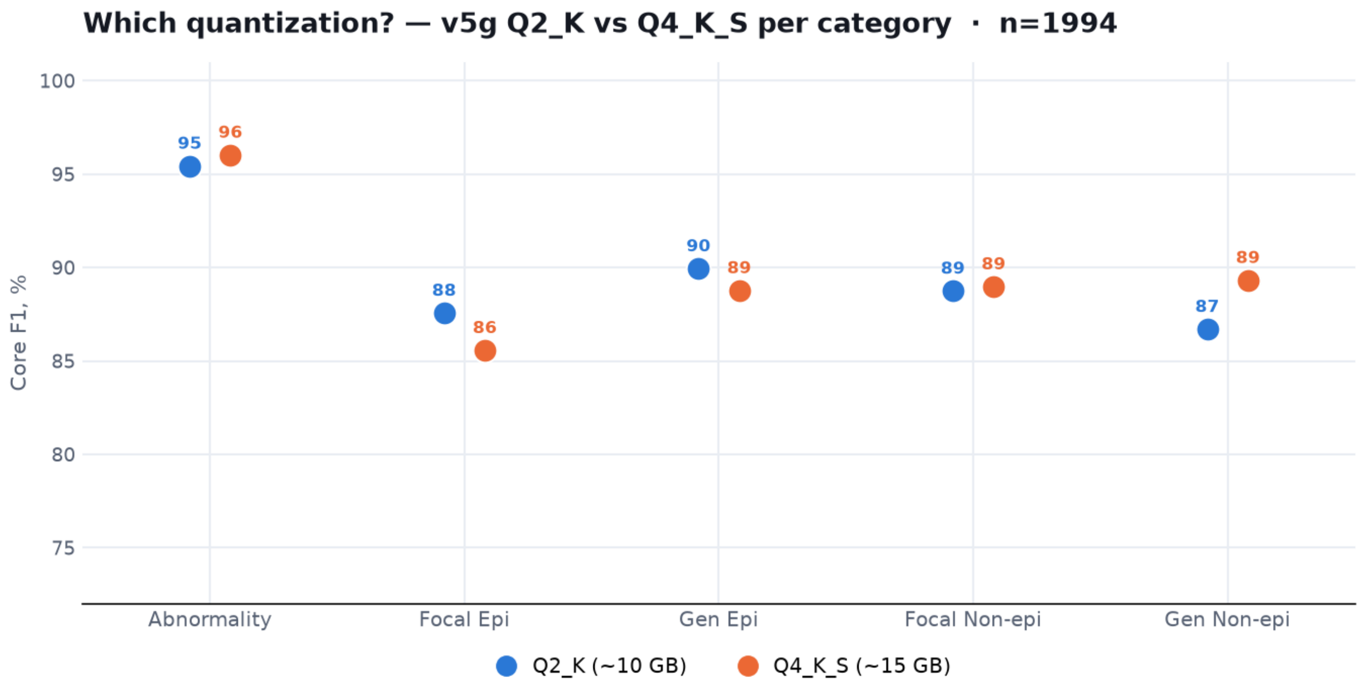

Figure 4: Q2 vs Q4 per category

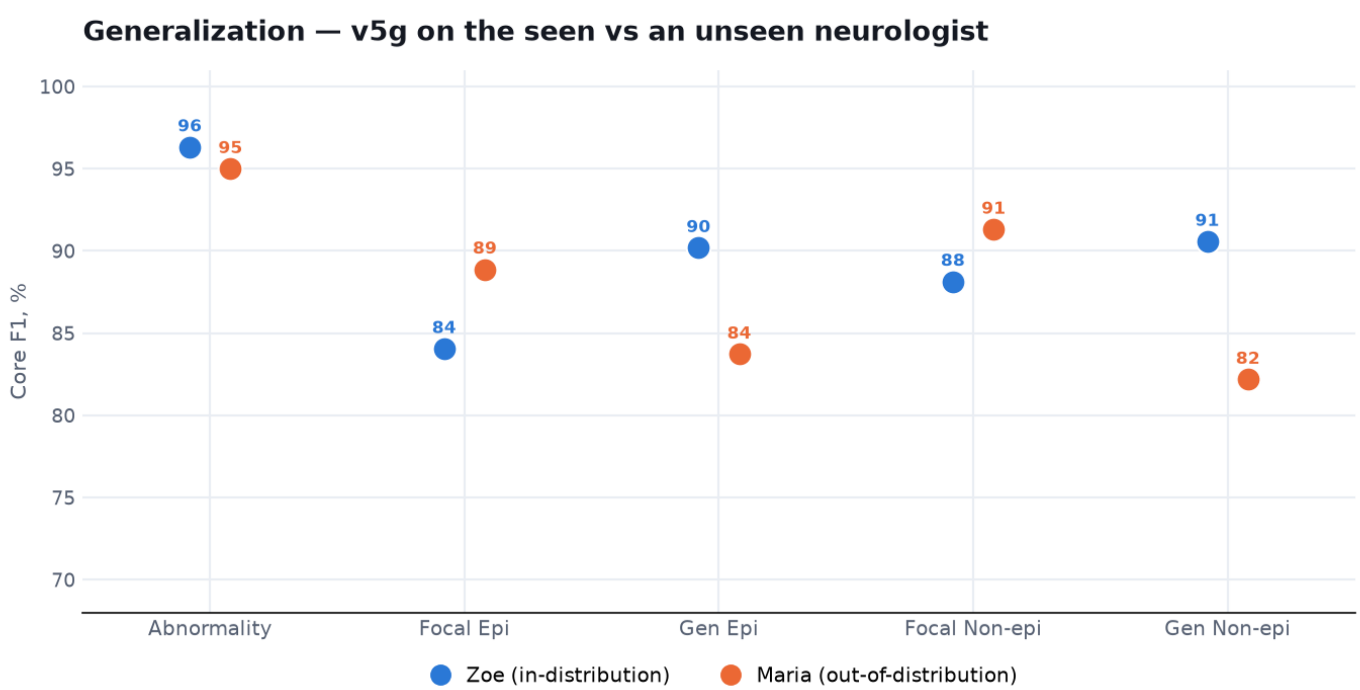

Figure 5: Generalization — Zoe vs Maria

How the metrics work (read before the tables)

Every label uses the **1–4 scale** (1 confident-no · 2 low-no · 3 low-yes · 4 confident-yes). We score with **F1 throughout**, at two strictness levels:

- **Core F1** — collapse to yes/no (1–2 = absent, 3–4 = present) and score how well "present" is detected. "3 vs 4" counts as correct (both mean present).
- **Certainty F1** — the exact 1–4 must match; "3 vs 4" is wrong (right direction, wrong confidence). The harder test — it shows whether the model captures *how sure* to be.

Why F1, not plain accuracy? The classes are very imbalanced (Focal/Gen Epi are present in only ~5% of reports). Plain accuracy would score ~98% just by always predicting "absent", hiding the failure to catch the rare findings. F1 = 2·TP / (2·TP + FP + FN) ignores the easy true-negatives and measures only how well the *present* class is actually caught (balancing precision and recall). It is the paper's core metric.

The one exception is Whole-report (All-5) — the share of reports where *all five* labels are correct. That is an exact-match count by nature (there is no F1 for a whole report), so it stays a plain percentage.

Full numbers (pooled n=1994)

Per-category values are **Core F1**; "Whole" is **All-5 accuracy** (share of reports with all five labels correct), vs LD and vs the held-out annotator SG.

Prompt	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	Whole vs LD	Whole vs SG
Mistral-7B	94.7	82.8	74.8	75.6	75.2	74.5	—
v1	96.8	80.4	88.5	88.0	84.0	83.9	82.9
v3	95.6	82.7	87.0	87.5	85.7	84.2	82.4
v3g	95.9	86.4	89.2	86.9	89.6	86.5	86.0
v5g Q2	95.4	**87.6**	**89.9**	88.8	86.7	**87.6**	85.6
v5g Q4	96.0	85.6	88.8	89.0	89.3	**87.6**	**86.2**
v7g	96.2	86.0	87.8	87.4	90.0	87.1	86.1
v8g	93.9	76.9	85.4	85.8	87.7	83.6	82.8
v10g	96.1	86.9	87.8	87.4	89.8	86.7	86.1
Human SG	98.0	85.7	87.5	90.8	90.0	89.8	—

(v9g omitted from the pooled table — it is much slower and did not finish the full run; on a fair same-case comparison it was 3–6 points **below** v5g. Values shown use Q4 for the grammar variants except where a quant is named.)

By dataset — Zoe (in-distribution, n=1495)

Prompt	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	Whole vs LD	Whole vs SG
Mistral-7B	96.1	84.0	72.5	76.0	79.0	75.3	—
v1	98.1	76.8	89.9	87.1	85.5	83.9	82.9
v3	96.9	79.5	87.4	87.6	87.8	85.1	82.9
v3g	96.4	85.3	90.8	85.5	91.0	86.1	85.5
v5g Q2	96.1	87.1	91.2	89.0	87.9	**87.8**	85.8
v5g Q4	96.3	84.1	90.2	88.1	90.6	87.2	85.6
v7g	96.6	84.7	88.9	86.5	91.7	86.8	85.9
v8g	93.7	72.5	85.9	83.9	88.9	82.2	81.7
v10g	96.5	85.9	88.9	86.7	90.9	86.2	85.4
Human (SG)	97.9	83.7	88.7	89.5	89.9	88.8	—

By dataset — Maria (out-of-distribution, n=499)

Prompt	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	Whole vs LD	Whole vs SG
Mistral-7B	89.8	80.6	83.7	74.5	54.0	71.9	—
v1	92.2	88.2	83.7	90.5	75.0	83.6	83.2
v3	90.8	90.3	85.7	87.2	73.4	81.6	81.0
v3g	94.5	88.9	83.7	90.8	81.6	87.8	87.6
v5g Q2	93.2	88.5	85.7	88.1	80.0	86.8	85.2
v5g Q4	95.0	88.9	83.7	91.3	82.2	**88.8**	**88.0**
v7g	94.8	88.9	83.7	90.1	80.6	87.8	86.8
v8g	94.7	87.0	83.7	91.1	80.8	87.8	86.4
v10g	94.8	88.9	83.7	89.4	83.6	88.2	88.0
Human (SG)	98.1	90.0	83.7	94.4	90.4	92.8	—

The model **generalizes**: on the unseen neurologist (Maria) the epileptiform and focal-non classes are as strong as on Zoe, and v5g still tracks the human. Gen Non-epi is the one class that is harder OOD (it is rarer on Maria, ~15%).

Certainty (exact-level) F1

The Full-numbers table above is **Core F1**. Here is the same, per category, at the stricter **Certainty** level — the exact 1–4 confidence must match. This is the harder test, and it is where our approach separates most from Mistral.

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non
Mistral-7B	72.4	41.4	69.2	44.8	52.2
v1	56.0	67.6	78.1	45.2	54.7
v3	59.8	69.2	74.6	50.5	56.5
v3g (Q4)	75.6	66.3	75.9	60.6	67.1
v5g (Q2)	65.8	76.8	83.6	59.3	60.1
v5g (Q4)	74.1	66.7	72.4	63.1	69.0
v7g (Q4)	77.6	66.0	74.5	62.2	69.6
v8g (Q4)	61.6	59.4	73.4	50.4	61.1
v10g (Q4)	75.1	67.7	77.6	62.0	71.5
Human (SG)	92.0	72.0	84.1	46.9	58.3

The rare **epileptiform** classes are the story: at the exact level Mistral collapses (Focal Epi 41) while our grammar prompts hold (66–77). On Abnormality the human is far ahead (92) — judging *how sure* to be about "abnormal" is the hardest thing to imitate. On the slowing classes even the human's certainty F1 is modest (Focal Non 47, Gen Non 58), and our models sit at or above it.

Certainty — Zoe (in-distribution, n=1495)

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non
Mistral-7B	72.5	42.7	65.5	41.4	55.0
v1	53.1	60.9	76.5	39.5	56.4
v3	56.7	63.0	71.5	44.6	59.0
v3g	77.3	64.7	73.7	64.3	71.8
v5g (Q2)	61.9	72.6	83.0	53.7	60.8
v5g (Q4)	74.7	63.8	70.6	65.8	72.2
v7g	78.6	61.3	73.2	63.8	73.0

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non
v8g	57.1	55.0	71.8	42.7	61.6
v10g	75.5	63.7	75.8	64.3	73.9
Human (SG)	91.5	69.8	84.2	42.5	58.9

Certainty — Maria (out-of-distribution, n=499)

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non
Mistral-7B	72.2	38.8	83.7	55.5	36.5
v1	66.7	82.4	83.7	61.5	44.1
v3	71.3	83.9	85.7	67.4	41.7
v3g	69.6	69.8	83.7	50.0	40.8
v5g (Q2)	79.2	85.2	85.7	75.1	56.3
v5g (Q4)	72.3	73.0	79.1	55.3	50.7
v7g	73.9	76.2	79.1	57.5	50.0
v8g	77.8	69.6	79.1	72.8	58.3
v10g	73.9	76.2	83.7	55.3	57.5
Human (SG)	93.5	76.7	83.7	59.1	54.8

Regenerate: `python -m analysis.f1_tables`.

Full matrix — every prompt × quantization

All variants at both model sizes (Core F1 vs LD, pooled n=1994). The grammar-enforced variants (…g) are the strong ones; the plain prompts (v1–v6) are shown for completeness.

Prompt	Quant	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	Whole
v1	Q2_K	96.8	80.4	88.5	88.0	84.0	83.9
v1	Q4_K_S	97.8	73.2	87.3	84.7	88.9	81.3
v2	Q2_K	97.6	63.6	88.1	85.9	82.4	79.0
v2	Q4_K_S	98.1	70.2	88.5	85.8	89.0	82.8
v3	Q2_K	95.6	82.7	87.0	87.5	85.7	84.2
v3	Q4_K_S	96.8	81.7	88.1	85.8	89.4	83.2
v4	Q2_K	96.0	82.5	86.8	87.3	86.6	83.4
v4	Q4_K_S	97.1	79.3	87.6	84.1	89.0	81.1
v5	Q2_K	92.6	82.5	89.0	88.8	86.8	82.8
v5	Q4_K_S	97.1	82.9	87.6	88.3	88.1	84.7
v6	Q2_K	93.6	83.8	88.0	83.6	85.1	82.0
v6	Q4_K_S	97.0	83.8	88.7	87.7	88.1	84.6
v3g	Q2_K	95.2	88.5	90.4	88.2	85.3	86.1
v3g	Q4_K_S	95.9	86.4	89.2	86.9	89.6	86.5
v5g	Q2_K	95.4	87.6	89.9	88.8	86.7	**87.6**
v5g	Q4_K_S	96.0	85.6	88.8	89.0	89.3	**87.6**
v7g	Q2_K	94.4	85.3	89.6	87.8	85.4	86.1
v7g	Q4_K_S	96.2	86.0	87.8	87.4	90.0	87.1
v8g	Q2_K	94.4	58.6	76.0	83.9	85.0	79.3
v8g	Q4_K_S	93.9	76.9	85.4	85.8	87.7	83.6
v10g	Q2_K	94.6	84.8	89.6	88.6	85.9	86.5
v10g	Q4_K_S	96.1	86.9	87.8	87.4	89.8	86.7

(v9g omitted — the reasoning-first run is much slower and did not finish; on a fair same-case comparison it was 3–6 points below v5g.)

What worked, what didn't

- **Won:** the combination **v5 prompt + grammar-enforced consistency (v5g)**. The grammar is the single biggest lever — it removes every overall/subtype contradiction *and*, by letting the model decide the parts before the whole, lifts the epileptiform F1.
- **Rejected — asking instead of enforcing (v6):** the model doesn't hold the consistency rule on its own.
- **Rejected — body-aware abnormality (v7):** the remaining missed-abnormal cases are genuinely ambiguous, not a prompt gap.
- **Rejected — simplification (v8):** the detailed focal-epileptiform exclusions are load-bearing; removing them re-introduces over-calling.
- **Rejected — reasoning-first (v9):** given latitude to deliberate, the model's unaided judgement still under-performs the explicit rules by 3–6 points.
- **Rejected — evidence calibration (v10):** ~1 point below v5g on both quants; the "prefer absent" rule slightly hurt the rare Focal Epi class on Q2.

The improvement **holds against the held-out annotator SG** (82.9% → 86.2%), so it is a real gain, not fitting to LD; the residual gap to the human is close to the level of human–human disagreement, which is a reason to validate carefully rather than a claim that nothing more is possible.

Per-prompt detail — core vs certainty, Q2 vs Q4

One chart per prompt, each overlaying both quantizations. The **filled dot** is Core F1 (present/absent), the **open circle** is strict Certainty F1 (exact 1–4 level); the line length is the drop when the exact level is required. All pooled over 1994 reports.

v5g (best):

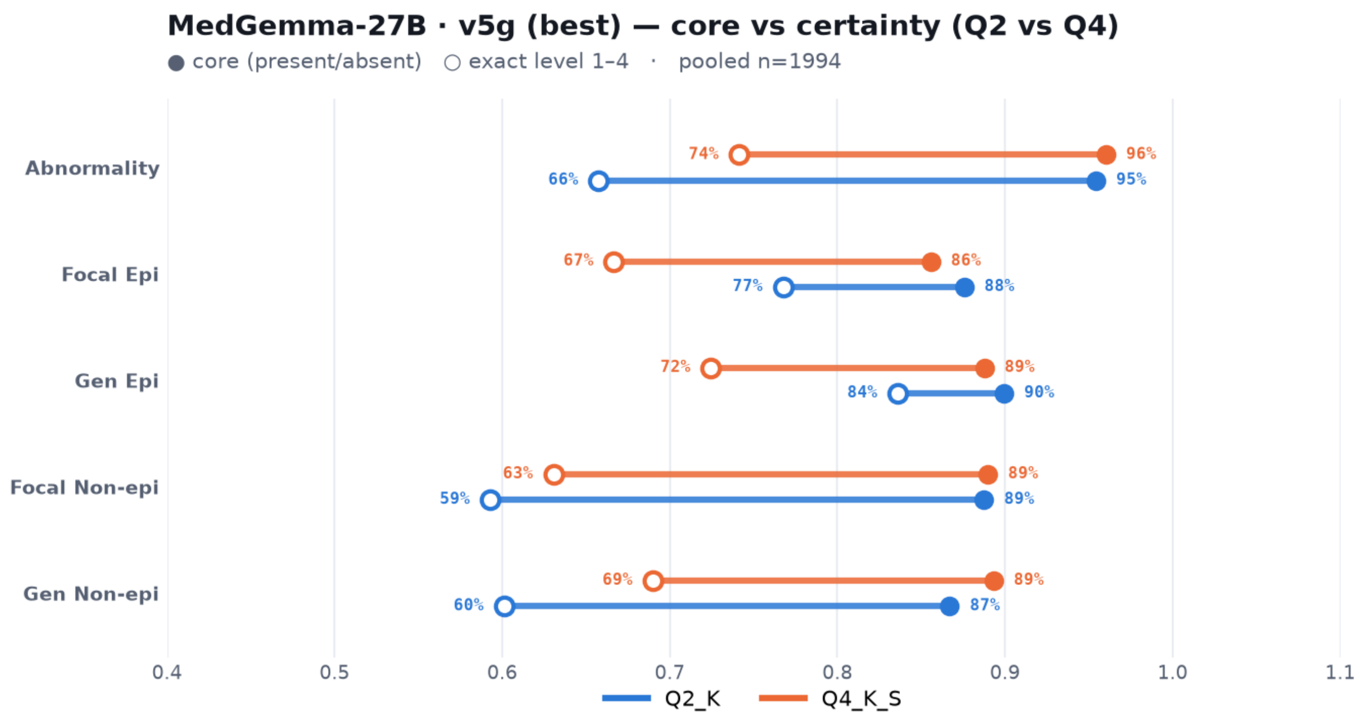

Figure 6: v5g core vs certainty

v3g:

v7g:

v8g (simplified):

v10g (calibrated):

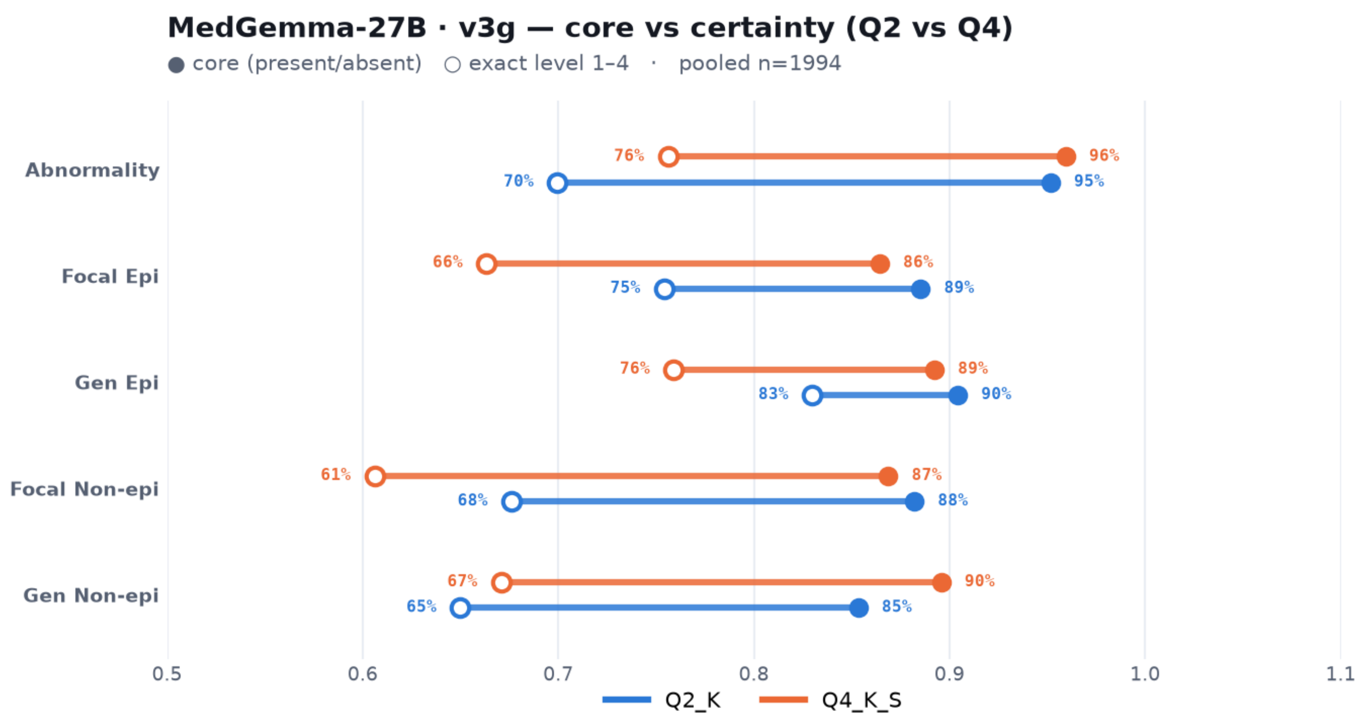

Figure 7: v3g core vs certainty

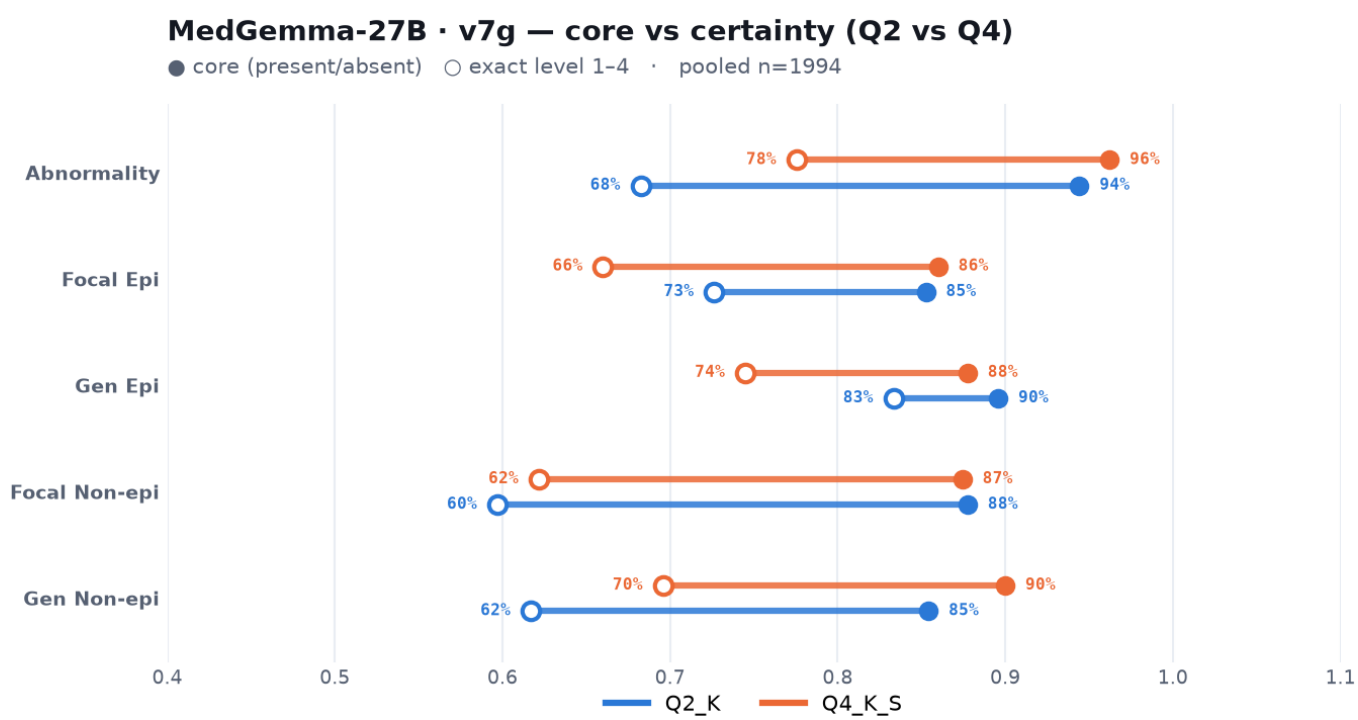

Figure 8: v7g core vs certainty

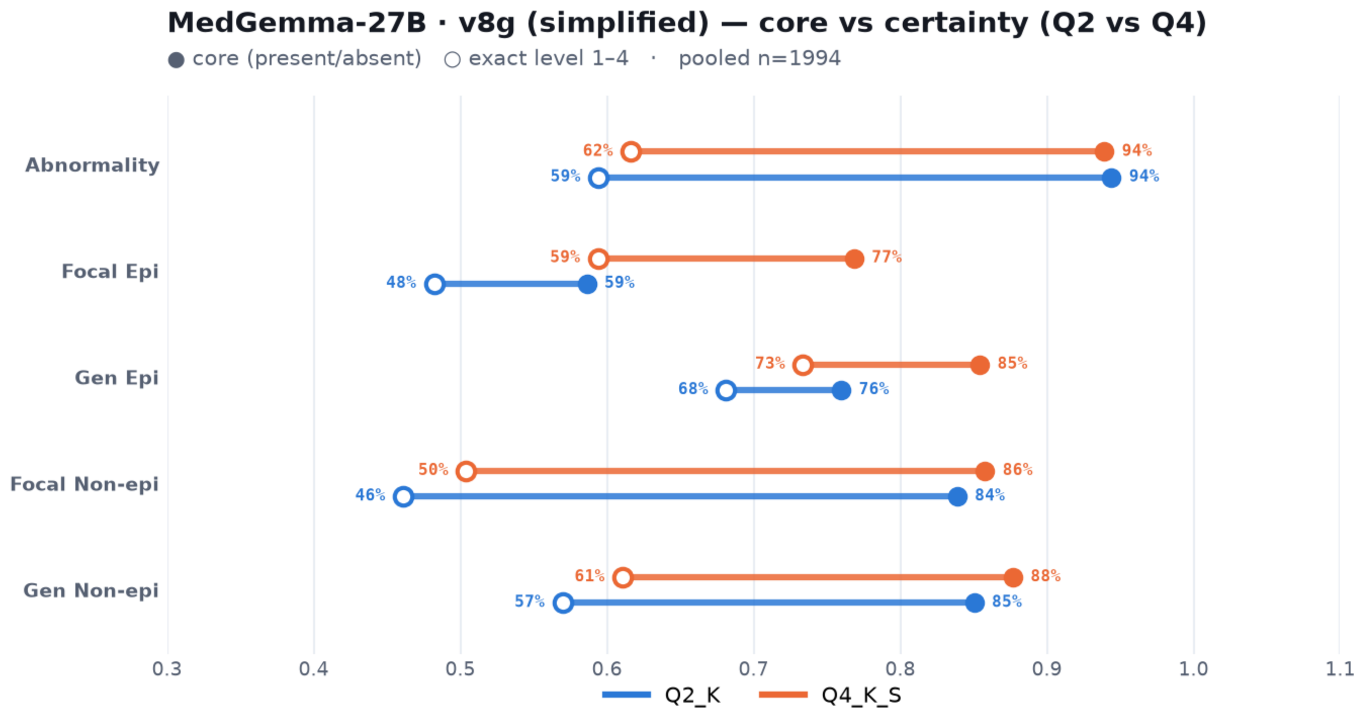

Figure 9: v8g core vs certainty

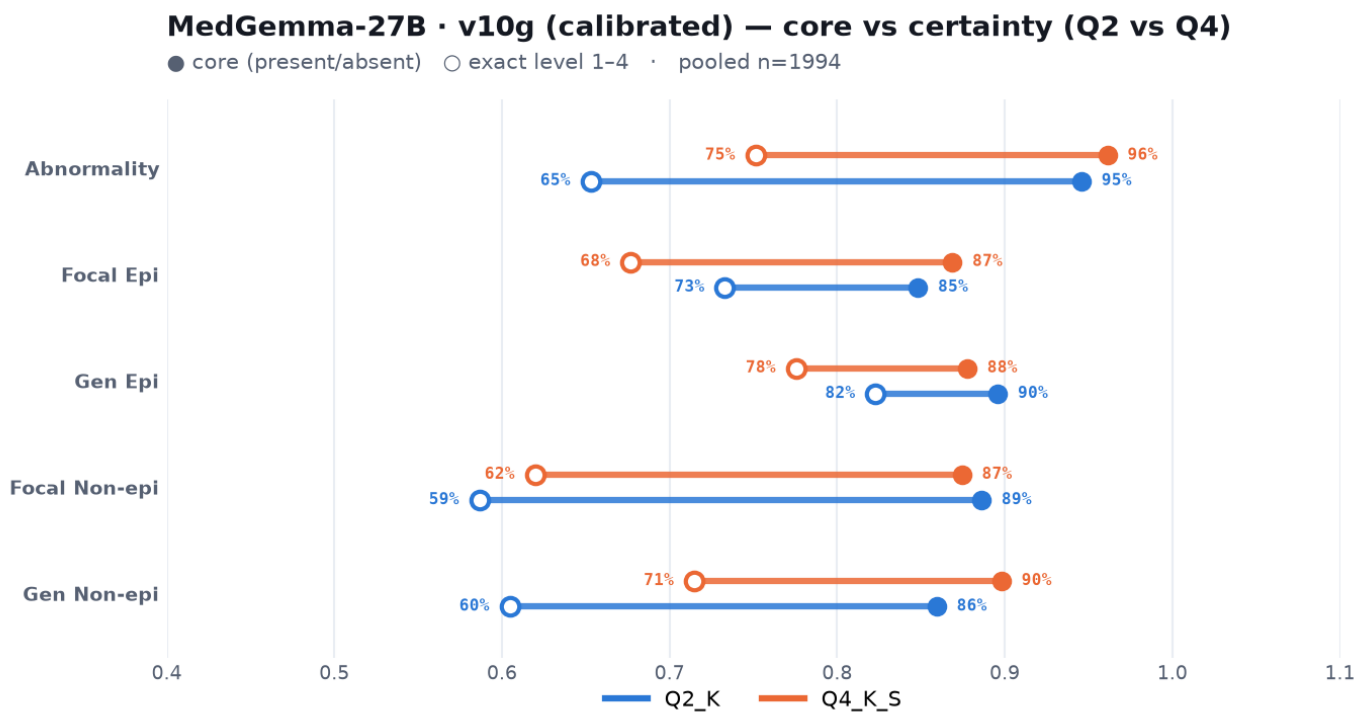

Figure 10: v10g core vs certainty

Mistral-7B (reference):

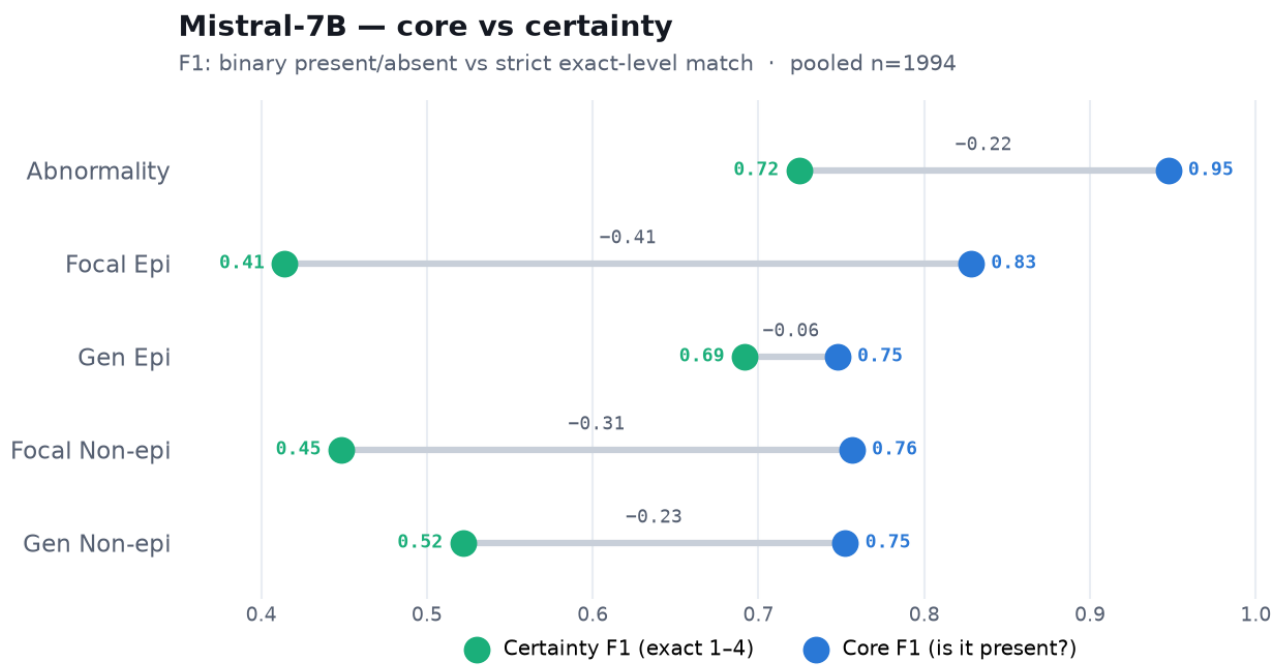

Figure 11: Mistral core vs certainty

Reproduce

```
pip install -e .
python -m analysis.make_tables          # the numbers tables
python -m analysis.plot_all_prompts     # every chart in this report -> reports/figures/
python -m analysis.validate_vs_sg       # cross-annotator validation
```

Per-run job IDs and hypotheses: `../experiments/`. Model: MedGemma-27B GGUF (Q2_K / Q4_K_S), llama.cpp grammar-constrained, temperature 0, 64-core CPU.